IP addressing & Subnetting

1. IPv4 Address Classes and Real-World Examples

Class	1st Octet Range	Default Subnet Mask	Primary Purpose / Allocation	Real-World Device or Service Example
Class A	$1\text{ – }126$	$255.0.0.0$ ($/8$)	Massive global organizations & ISPs ($16\text{M}+$ hosts per network).	**Apple / IBM Corporate Backbones** or Google Public DNS (8.8.8.8).
Class B	$128\text{ – }191$	$255.255.0.0$ ($/16$)	Medium-to-large enterprise networks & universities ($65,534$ hosts).	**University Campus Networks** (e.g., student lab subnets on 172.16.0.0).
Class C	$192\text{ – }223$	$255.255.255.0$ ($/24$)	Small business LANs and home networks (254 hosts).	**Home Wi-Fi Router / Smart TV** (e.g., 192.168.1.1 gateway).
Class D	$224\text{ – }239$	*None (No Host/Net split)*	Multicast streaming (one-to-many packet distribution).	**IPL Live Video Streaming / OSPF Routing** (e.g., 224.0.0.5 for router updates).
Class E	$240\text{ – }255$	*None (Experimental)*	Reserved by IETF for research, testing, and future use.	**Internet Research Labs** (e.g., test protocols running on 240.0.0.1).

(Note: The $127.x.x.x$ network is reserved for loopback and diagnostic testing, e.g., 127.0.0.1).

2. IP Calculation for 192.168.10.45 / 255.255.255.0

Step-by-Step Working:

1. **Analyze the Subnet Mask:**
    - IP Address: 192.168.10.45
    - Subnet Mask: 255.255.255.0 (CIDR: /24)
    - In binary: 11111111 . 11111111 . 11111111 . 00000000
    - The first 3 octets (24 bits) represent the **Network portion**, and the 4th octet (8 bits) represents the **Host portion**.
2. **Calculate Network Address (Host bits set to all 0s):**
    - Perform bitwise AND between IP and Mask:

$$\text{192.168.10.45 AND 255.255.255.0} = \mathbf{192.168.10.0}$$

3. **Calculate Broadcast Address (Host bits set to all 1s):**
    - Change all 8 host bits in the 4th octet to binary 1 (255 in decimal):

$$\text{Broadcast Address} = \mathbf{192.168.10.255}$$

4. **Determine the Valid Host Range:**
    - First Usable Host = Network Address $+ 1 = \mathbf{192.168.10.1}$
    - Last Usable Host = Broadcast Address $- 1 = \mathbf{192.168.10.254}$

Final Summary:

- **Network Address:** 192.168.10.0
- **Broadcast Address:** 192.168.10.255
- **Valid Host Address Range:** 192.168.10.1 to 192.168.10.254 (Total 254 usable hosts)

3. Subnetting Plan for Zomato-Style Startup ($10.0.0.0/24$)

Subnet Calculations:

- **Starting Network:** $10.0.0.0/24$ (256 total addresses).
- **Number of subnets needed:** 4.
- **Bits borrowed:** $2^n \ge 4 \implies n = 2\text{ bits}$.
- **New Prefix Length:** $/24 + 2 = \mathbf{/26}$.
- **New Subnet Mask:** Borrowing 2 bits gives $128 + 64 = \mathbf{192} \implies \mathbf{255.255.255.192}$.
- **Block Size (Increment):** $256 - 192 = \mathbf{64}$.
- **Usable Hosts per Subnet:** $2^{(32 - 26)} - 2 = 64 - 2 = \mathbf{62\text{ hosts}}$.

Subnetting Allocation Table:

Group Name	Network Address	Subnet Mask	Usable Host Range	Broadcast Address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255

4. College Wi-Fi Scenario ($172.16.0.0/16$ for 20+ Subnets)

Calculation Steps:

1. **Determine Borrowed Subnet Bits (n):**
    - Formula: $2^n \ge \text{Number of required subnets}$
    - For 20 subnets:
        - $2^4 = 16$ (insufficient)
        - $2^5 = 32 \ge 20$ (satisfies requirement)
    - Therefore, we borrow **5 bits** from the host portion.
2. **Calculate New Subnet Mask:**
    - Original CIDR: $/16$
    - New CIDR: $/16 + 5 = \mathbf{/21}$
    - 3rd Octet Binary: 5 network bits followed by 3 host bits $\rightarrow `11111000` = 128 + 64 + 32 + 16 + 8 = \mathbf{248}$
    - **New Subnet Mask:** $\mathbf{255.255.248.0}$
3. **Calculate Available Hosts per Subnet:**
    - Total host bits remaining (h) = $32 - 21 = \mathbf{11\text{ bits}}$.
    - Formula for usable hosts: $2^h - 2$
    - Calculation: $2^{11} - 2 = 2048 - 2 = \mathbf{2,046\text{ usable hosts}}$.

Final Answers:

- **New Subnet Mask: $255.255.248.0$** (or **/21**)
- **Available Hosts per Subnet: $2,046$ usable hosts** (across up to 32 distinct building subnets).

5. Explanation for Invalid IP 192.168.1.300

- **The Problem:**
    - An IPv4 address consists of 4 octets separated by dots, where each octet is represented by **8 binary bits**.
    - The maximum value 8 bits can store is $2^8 - 1 = \mathbf{255}$ (range is strictly $0\text{ to }255$).
    - The value **300** exceeds this 8-bit limit, making 192.168.1.300 a syntactically invalid IPv4 address that operating systems and network stacks will reject.
- **What a Valid IP in that Subnet Looks Like:**
    - In a standard /24 home/office subnet (192.168.1.0/24), valid usable host addresses must fall strictly between:

$$\mathbf{192.168.1.1} \quad \text{and} \quad \mathbf{192.168.1.254}$$

- *Example fix:* Assign an unused address within range, such as **192.168.1.30** or **192.168.1.50**, with subnet mask **255.255.255.0** and default gateway **192.168.1.1**.